

Assignment No. 2

MTH501 (Spring 2021)

Total Marks: 10
Due Date: 18-02-2022

DON'T MISS THESE Important instructions before attempting the solution of this assignment:

- To solve this assignment, you should have good command over **23-31 lectures**.
- Upload assignments properly through LMS, No Assignment will be accepted through email.
- Write your ID on the top of your solution file.
- Don't use colorful back grounds in your solution files.
- Use Math Type or Equation Editor etc. for mathematical symbols if needed.
- **You should remember that if the solution files of some students are found same (copied) then we will reward zero marks to all those students.**
- **Make solution by yourself and protect your work from other students, otherwise both original and copied assignments will be awarded zero marks.**
- **Also remember that you are supposed to submit your assignment in Word format, any other format like scanned images etc. will not be accepted and be awarded zero marks.**
- **You are to attempt both questions however, only one question will be graded as per the choice of your section in charge.**

Question 1

Find the Eigenvalues and Eigenvectors for the matrix:

$$A = \begin{bmatrix} 5 & -10 & -5 \\ 2 & 14 & 2 \\ -4 & -8 & 6 \end{bmatrix}$$